INCH-POUND

MIL-B-44474
20 September 1993

MILITARY SPECIFICATION

BEEF WITH MUSHROOMS IN SAUCE, THERMOSTABILIZED,

FOR MEAL, READY-TO-EAT

This specification is approved for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers thermostabilized beef with mushrooms in sauce in flexible pouches for use as a component of the Meal, Ready-to-Eat, Individual.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5018 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8940

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SPECIFICATION

MILITARY

MIL-P-44073 - Packaging and Thermoprocessing of Foods in Flexible Pouches

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Primary Drinking Water Regulations

(Copies are available from the Office of Drinking Water, Environmental Protection Agency, WH550D, 401 M Street, S.W., Washington, DC 20460.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

General Specifications for Dairy Plants Approved for USDA Inspection and Grading Service (7 CFR Part 58)

(Copies are available from the Dairy Standardization Branch, Dairy Division, Agricultural Marketing Service, U.S. Department of Agriculture, Room 2750, South Building, P.O. Box 96456, Washington, DC 20090-6456.)

Institutional Meat Purchase Specifications for Fresh Beef, Series 100

(Copies are available from the Chief, Livestock and Meat Standardization Branch, Livestock and Seed Division, Agricultural Marketing Service, U.S. Department of Agriculture, Room 2603, South Building, P.O. Box 96456, Washington, DC 20090-6456.)

Meat and Poultry Inspection Regulations (9 CFR Part 301-391)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

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U.S. Standards for Grades of Canned Mushrooms (7 CFR Part 52)

(Copies are available from the Chief, Processed Products Branch, Fruit and Vegetable Division, Agricultural Marketing Service, U.S. Department of Agriculture, Room 0709, South Building, P.O. Box 96456, Washington, DC 20090-6456.)

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES (HHS), U.S. FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder (21 CFR Parts 1-199)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.1).

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

AMERICAN DEHYDRATED ONION AND GARLIC ASSOCIATION (ADOGA)

Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products

(Application for copies should be addressed to the American Dehydrated Onion and Garlic Association, One Maritime Plaza, 23rd Floor, San Francisco, CA 94111.)

AOAC INTERNATIONAL

Official Methods of Analysis of the AOAC

(Application for copies should be addressed to the AOAC International, 2200 Wilson Boulevard, Suite 400, Arlington, VA 22201-3301.)

NATIONAL ACADEMY OF SCIENCES

Food Chemicals Codex

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(Application for copies should be addressed to National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.1), a sample shall be subjected to first article inspection (see 6.2) in accordance with 4.4.

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Beef. The beef shall be from steers, heifers, or cows and shall be derived from any one or any combination of the following USDA Certified Institutional Meat Purchase Specifications (IMPS) cuts: Item No. 114 (shoulder, clod), Item No. 161 (round, shank-off, boneless), Item No. 163 (round, shank-off, three-way, boneless), Item No. 165 (round, rump, shank-off, boneless), Item No. 167A (round, knuckle, peeled), Item No. 169 (top round), Item No. 170A (bottom round, heel out) and Item No. 171B (outside round). All beef shall be certified by a USDA Agricultural Marketing Service (AMS) agent for condition of the product and compliance with the IMPS Quality Assurance Provisions.

NOTE: Certified IMPS cuts may be further trimmed (see 3.3.1) and broken down into smaller cuts after certification as requested by the purchaser. Boxes used for shipment of the further trimmed and broken down cuts shall be labeled, "Beef for further processing as specified in MIL-B-44474". The certificate shall be noted as such and shall accompany the shipment.

3.2.1.1 Handling and storage. Handling and storage of the boned and trimmed beef, prior to processing into the finished product (see 3.3.1), shall be in accordance with the following requirements:

a. Beef processed on the day of initial certification shall be maintained in the temperature range of 28°F to 50°F, inclusive.

b. Holding in the fresh-chilled state for not more than four days after initial certification is permitted, provided that the beef is maintained in the temperature range of 28°F to 40°F, inclusive.

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c. Holding in the frozen state for not more than 180 days after initial certification is permitted, provided that the beef is:

- frozen to 0°F or lower within 72 hours after initial certification
- stored at 0°F or lower.
- protected from freezer deterioration and damage
- stored in containers that are adequate to maintain product excellence.

d. Tempering/thawing of frozen beef shall comply with the following:

- microwave tempering/thawing of frozen beef is permitted provided excellent condition of the beef, as determined by the USDA, is maintained
- the microwave tempering/thawing process shall not exceed 48 hours from time of removal of the beef from the freezer to further preparation
- regardless of the process used, frozen beef shall be tempered/thawed to a temperature not to exceed 40°F
- when a tempering/thawing process other than microwave is used, the time from the start of the tempering/thawing process (removal from freezer) to further preparation shall not exceed 96 hours
- beef which has been tempered/thawed shall not be refrozen.

3.2.2 Beef, cooked, frozen. Frozen cooked beef in the log or dice form shall have been prepared from beef that complies with the requirements of 3.2.1 and 3.2.1.1. The beef log or dices shall be packaged and sealed in USDA approved food packaging material and shall be frozen to an internal temperature of 0°F or below within 72 hours. The beef logs or dices shall be certified by a USDA Agricultural Marketing Service (AMS) agent for condition of the product, the initial freezing (in-storage) date and the above stated requirements. The packaged, frozen, cooked beef logs or dices shall have been held at an internal temperature of 0°F or below for a period not to exceed 120 days prior to product preparation (see 3.3.5).

3.2.3 Water. Water used for formulation, ice making, and washing shall conform to the National Primary Drinking Water Regulations.

3.2.4 Mushrooms, canned. Canned mushrooms shall be stems and pieces or sliced whole style, U.S. Grade B or better in accordance with the U.S. Standards for Grades of Canned Mushrooms (7 CFR Part 52).

3.2.5 Cream, dry. Dry cream shall comply with the definition and standard of identity for dry cream except it shall contain 70 percent to 75 percent, by weight, of milkfat (21 CFR Part 131). The product shall not contain added nutritive, carbohydrate sweeteners. The dry cream shall be manufactured in accordance with the General Specifications for Dairy Plants Approved for USDA Inspection and Grading Service (7 CFR Part 58).

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3.2.6 Onions, dehydrated, chopped. Dehydrated chopped onions shall be Fancy grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products.

3.2.7 Starch, food, modified, high opacity. The high opacity starch shall be white, odorless, finely pulverized, modified waxy maize food starch for use in thermostabilized foods. The modified starch shall demonstrate initial viscosity development in the temperature range of 140°F to 170°F and be fully hydrated at common retort temperatures. The starch shall resist breakdown at low pH, under shear stress, and under conditions of cold storage. The cooked slurry prepared from the starch shall be bland with essentially no cereal or starch taste.

3.2.8 Salt. Salt shall be noniodized, white, refined, sodium chloride, with or without anticaking agents and shall comply with the purity standards for sodium chloride of the Food Chemicals Codex.

3.2.9 Mustard flour. Mustard flour shall be a bright yellow powder from a blend derived from the endosperm of the seed of Brassica hirta and Brassica juncea. Mustard flour shall contain not less than 0.4 mL volatile oil per 100 grams of mustard flavor and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 60 sieve.

3.2.10 Lecithin. Lecithin shall comply with the Food Chemicals Codex.

3.2.11 Flavoring, natural, braised beef. The natural braised beef flavoring shall be a tan to light brown, free flowing powder. A one percent water solution shall demonstrate typical braised beef flavor and aroma notes. The moisture content of the powder shall not exceed 8.0 percent. The powder shall have a standard plate count not greater than 10,000/gram; a yeast and mold count not greater than 100/gram; an E. coli count less than 3/gram; Coagulase (+) S. aureus not greater than 100/gram and shall be Salmonella negative (see 6.3.1).

3.2.12 Pepper, black, ground. Ground black pepper shall have been ground from the deep brown to black, deep-set, wrinkled, immature berries of Piper nigrum L. The ground pepper shall have a characteristic, penetrating odor, a hot, biting, pungent flavor and a light gray to speckled black-gray color. The ground pepper shall contain not less than 2.0 mL of volatile oil per 100 grams of ground black pepper and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 16 sieve.

3.2.13 Allspice, ground. Ground allspice shall be derived from dried, nearly ripe fruit of Pimenta officinalis Lindl and shall possess a fragrant, clove-like aroma, strongly aromatic, pungent, clove-like flavor and a dark reddish-brown color. Allspice shall contain not less than 3.0 mL of volatile oil per 100 grams of ground allspice with 80.0 percent eugenol as a principal constituent. A minimum of 95 percent, by weight, shall pass through a U.S. Standard No. 25 sieve.

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3.2.14 Garlic powder. Garlic powder shall be Fancy grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Product.

3.2.15 Preblended spice and seasoning mixture. Preblended spices and seasonings may be used. The spices and seasonings in the mixture shall comply with the requirements of this specification. The containers used for the spice and seasoning blend shall be labeled with each ingredient and the percentage of each ingredient in the blend. The ingredients shall be in the same proportions as specified in the ingredient formula.

3.2.16 Starch, waxy maize, modified, filling and processing aid. The starch shall be a white to off-white, odorless, finely pulverized powder suitable for use as a filling and processing aid in thermostabilized foods. When hydrated at temperatures of 165°F to 200°F, the starch shall develop a paste. The initial viscosity of the paste shall be retained under conditions of moderate shear and prolonged holding times. During retorting at temperatures in excess of 200°F, the starch shall undergo extensive to complete breakdown of the initial viscosity. The starch shall be bland and shall demonstrate easy dispersibility.

3.2.17 Sodium tripolyphosphate. Sodium tripolyphosphate shall comply with the Food Chemicals Codex.

3.3 Preparation and processing. Processing shall be on a continuous basis.

3.3.1 Beef preparation. Boned and trimmed beef meeting the requirements of 3.2.1 shall be further trimmed, if necessary, so as to assure compliance with finished product requirements and shall be processed as follows:

a. The whole beef cuts shall be mechanically diced with dicer settings at 5/8 by 5/8 by 5/8 inch dimensions (± 1/16 inch for each dimension).

b. The raw beef dices shall be held for not more than 24 hours at an internal temperature of 28°F to 40°F from time of dicing to beginning of cooking process.

c. The beef dices shall be cooked in water, or by other commercially acceptable cooking method, until no raw portions are visible. The cooked beef dices shall be thoroughly drained.

d. The cooked beef dices shall be held for not more than 48 hours at an internal temperature of 28°F to 40°F until filled into the pouches or, if stored in the frozen form, until packaged and sealed (see 3.2.2).

3.3.1.1 Beef log preparation and processing. As an alternative to preparing and dicing whole beef cuts, the dices may be prepared from beef logs. The beef logs shall be prepared and processed as follows:

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Ingredients	Percent by weight
Beef	95.75
Ice or ice water	3.00
Salt	1.00
Sodium tripolyphosphate	0.25

a. The beef shall be mechanically ground through a grinder plate having kidney-shaped openings measuring not less than 2 inches in the shortest dimension. The grinder shall be equipped with a two-blade knife.

b. The ground beef shall be mechanically mixed with the ice or ice water, sodium tripolyphosphate, and the salt. The mixing shall continue until the mixture exhibits a sticky (tacky) consistency. The time from grinding to mixing shall not exceed four hours. The ground beef shall be maintained at an internal temperature of 28°F to 40°F during this time period.

c. The mixed beef shall be immediately stuffed into casings of a type to accommodate the cooking method. Alternatively, the mixed beef may be formed in stainless steel molds. The formed beef logs shall be in the cooking process within 24 hours after forming.

d. The beef logs shall be cooked in a cook house (smokehouse without smoke) or by other commercially acceptable methods to provide a product meeting the finished product drained weight and other requirements. Water cooking is permitted only for the beef logs in casings provided they are cooked in water until the internal temperature of the log reaches 150°F to 160°F.

e. Immediately after completion of the cooking process the cooked beef logs shall be placed in an ice water bath, or cooled by other commercially acceptable rapid cooling methods. During cooling, the internal temperature of the beef log shall not remain between 130°F and 80°F for more than 1-1/2 hours nor between 80°F and 40°F for more than five hours. The cooled logs shall be held for not more than 48 hours at an internal temperature of 28°F to 40°F prior to dicing or packaging for frozen storage (see 3.2.2).

f. The cooked beef logs shall be mechanically diced with dicer settings at 5/8 by 5/8 by 5/8 inch dimensions (± 1/16 inch for each dimension). The diced beef shall be held not longer than 48 hours in the temperature range of 28°F to 40°F until preparation of the finished product or, if stored in the frozen form, until packaged and sealed (see 3.2.2). Dices prepared from previously frozen logs shall not be frozen.

3.3.2 Preparation for the canned mushrooms. The canned mushrooms shall be thoroughly drained prior to use in product preparation (see 3.3.5).

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3.3.3 Sauce preparation. The sauce shall be formulated and prepared as follows:

Ingredients	Percent by weight
Water	79.25
Cream, dry	8.53
Onions, chopped, dehydrated	5.81
Starch, food, modified, high-opacity	2.89
Salt 1/	1.54
Mustard flour	0.97
Lecithin	0.48
Flavoring, natural, braised beef	0.32
Pepper, black, ground	0.09
Allspice, ground	0.06
Garlic powder	0.06

1/ The total amount of salt in the sauce formula shall be adjusted, as necessary, to produce a product that complies with the finished product salt requirements (see 3.6).

NOTE: The following sauce preparation procedures were used in the development of this product. Alternative procedures may be used provided all finished product requirements are met. When alternative procedures are used, the time and temperature requirements specified for the prepared sauce are still applicable.

a. The slurry shall be made using the starch and part of the water.

b. The remaining ingredients shall be mixed together and heated to not less than 140°F.

c. The starch slurry shall be added and the final mixture heated to a temperature of 150°F to 180°F and held in this temperature range for five minutes.

d. The volume of the final mixture shall be adjusted with water to compensate for evaporation loss during heating and holding.

e. The prepared sauce shall be immediately combined with the cooked beef dices and mushrooms (see 3.3.5).

3.3.4 Filling and processing aid, starch preparation. When a filling and processing aid is used, a solution (slurry) shall be prepared with the filling and processing aid starch and part of the sauce formulation water in a concentration adequate to hold ingredients in suspension for uniform distribution during mixing and pumping of the product.

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3.3.5 Product preparation. The diced beef, mushrooms and sauce shall be combined in the following approximate proportions, using either a single-stage or multistage filling operation. The percent by weight of ingredients may be adjusted if necessary to ensure compliance with the finished product drained weight requirements.

Ingredient	Percent by weight
Beef, diced, cooked	43.00 1/
Sauce	47.00
Mushrooms	10.00

1/ Based on an approximate 80 percent yield.

The prepared product shall be maintained in the temperature range of 150°F to 180°F and filled into the pouches within one hour after preparation

3.4 Pouch filling and sealing. Each pouch (see 5.1) shall be filled with product such as to conform to the finished product requirements and to the following requirements:

a. Each pouch shall be filled and sealed in accordance with the pouch filling and sealing requirements specified for class 1 of MIL-P-44073.

b. The temperature of the product at time of filling shall be 150°F to 180°F.

c. Each filled and sealed pouch shall be in the retort process within two hours after sealing.

3.5 Pouch thermoprocessing. The filled and sealed pouches shall be thermostabilized by retorting until a sterilization value (F_o) of not less than 6 has been achieved. The thermoprocessing operation shall be in compliance with the thermoprocessing requirements for class 1 of MIL-P-44073.

3.6 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material such as, but not limited to, dirt, insect parts, hair, wood, glass, or metal.

b. There shall be no foreign odor or flavor such as, but not limited to, burnt, scorched, stale, sour, rancid, or moldy.

c. There shall be no color foreign to the product.

d. The average net weight shall be not less than 8.0 ounces.

e. No individual pouch shall contain less than 7.5 ounces of product.

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f. The total weight of cartilage, coarse connective tissue, sections of tendons, or ligaments, and glandular material, collectively, shall be not more than 0.35 ounce.

g. No individual pouch shall contain a bone piece measuring 0.3 inch or more in any dimension.

h. Texture of the beef dices shall not be dry, rubbery, or mushy.

i. Texture of the mushrooms shall not be mushy, hard, fibrous, or tough.

j. The sauce shall be a cream to light tan color.

k. Sauce consistency shall be smooth, without lumps, and shall not be excessively thick or thin.

l. The product shall show no evidence of excessive heating (materially darkened or scorched).

m. No individual pouch shall contain less than 2.0 ounces of drained beef.

n. No individual pouch shall contain less than 0.5 ounces of drained mushrooms.

o. Of the 2.0 ounces drained weight of beef, not less than 1.5 ounces shall be whole intact beef dices.

p. The average drained weight of beef shall be not less than 2.5 ounces.

q. The average drained weight of mushrooms shall be not less than 0.7 ounces.

r. The average fat content of the finished product shall be not greater than 8.0 percent and no individual pouch shall have a fat content greater than 10.0 percent.

s. The salt content of the product in any individual pouch shall be not greater than 1.3 percent nor less than 0.8 percent.

3.6.1 Palatability. The finished product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.7 Plant qualification. The beef component and the finished product shall originate and be produced, processed, and stored in plants regularly operating under the Meat and Poultry Inspection Regulations of the U.S. Department of Agriculture.

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3.8 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this specification. The contractor shall ensure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of inspection and supervision necessary to ensure compliance with the requirements of this specification.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this specification and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this specification or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material examination. In accordance with 4.1, components and materials shall be examined in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, and compliance shall be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods, such as the standard methods described in the Official Methods of Analysis of the AOAC and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the requirements. Any nonconformance to an identity, condition, or other

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requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, filling, sealing, and packaging requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

4.5.3 Filled and sealed pouch examination. Examination of the filled and sealed pouches shall be in accordance with the quality assurance provisions of MIL-P-44073.

4.5.4 Net weight examination. The net weight of the filled and sealed pouches shall be determined by weighing each sample unit on a suitable scale tared with a representative empty pouch. Any individual net weight of less than 7.5 ounces shall be scored as a minor defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-3 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5. Results shall be reported to the nearest 0.1 ounce. In addition, the lot shall be rejected if the sample average net weight is less than 8.0 ounces.

4.5.5 Product and drained weight examination.

4.5.5.1 Product examination. The filled and sealed sample pouches shall be held for a minimum of 72 hours at room temperature (65°F to 75°F) after completion of the thermoprocessing operation. The pouches shall be heated in 180°F to 190°F water for 10 minutes, opened, and inspected. The lot size shall be expressed in pouches. The sample unit shall be the contents of one pouch. The sample size shall be as indicated by the sampling plan specified in table I. The contents shall be examined for the defects indicated in table II.

TABLE I. Double sampling plan for product examination 1/

Lot size (pouches)	Sample size (pouches)	Cumulative sample	Acceptance number	Rejection number
0 to 500,000	20	—	0	2
	20	40	1	2

1/ a. If no defects are found in the first sample, the lot shall be accepted.

b. If the number of defects found in the first sample equals or exceeds the rejection number, the lot shall be rejected.

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c. If the number of defects found in the first sample exceeds the acceptance number but is less than the rejection number, the second sample shall be examined. Defects found in the first and second sample shall be combined and if the number of defects in the cumulative sample equals or exceeds the rejection number, the lot shall be rejected.

TABLE II. Product defects 1/ 2/

Category	Defect
Major	
101	The total weight of cartilage, coarse connective tissue, section of tendons or ligaments, and glandular material, collectively, in any individual pouch is more than 0.35 ounces
102	Presence of a bone piece measuring 0.3 inch or more in any dimension
103	Texture of the beef dices is dry, rubbery, or mushy
104	Texture of the mushrooms is mushy, hard, fibrous, or tough
105	Sauce not a cream to light tan color
106	Sauce consistency is lumpy or is excessively thick or thin
107	Product shows evidence of excess heating (materially darkened or scorched)

1/ The presence of any foreign material (for example, dirt, insect parts, hair, wood, glass, or metal), foreign odor or flavor (for example, burnt, scorched, moldy, rancid, sour, stale), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample in palatability and overall appearance shall be cause for rejection of the lot (see 3.6.1).

4.5.5.2 Drained weight examination. The filled and sealed sample pouches shall be held for a minimum of 72 hours at room temperature (65°F to 75°F) after completion of the thermoprocessing operation. The pouches shall be heated in 180°F to 190°F water for approximately 10 minutes, opened, and inspected. The lot size shall be expressed in pouches. The sample unit shall be the contents of one pouch. The sample size shall be as

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indicated by the double sampling plan specified in table III. The drained weights shall be determined and examined for the defects listed in table IV.

TABLE III. Double sampling plan for drained weight examination 1/

Lot size (pouches)	Sample size (pouches)	Cumulative sample	Acceptance number	Rejection number
0 to 500,000	20	--	0	2
	20	40	1	2

1/ a. If no defects are found in the first sample, the lot shall be accepted.

b. If the number of defects found in the first sample equals or exceeds the rejection number, the lot shall be rejected.

c. If the number of defects found in the first sample exceeds the acceptance number but is less than the rejection number, the second sample shall be examined. Defects found in the first and second sample shall be combined, and if the number of defects in the cumulative sample equals or exceeds the rejection number, the lot shall be rejected.

TABLE IV. Drained weight defect 1/

Category	Defect
Major	
101	Drained weight of beef in a pouch is less than 2.0 ounces 2/
102	Drained weight of mushrooms in a pouch is less than 0.5 ounces 3/
103	Of the 2.0 ounces drained weight of beef, less than 1.5 ounces are whole intact beef dices 4/

1/ To determine drained weight, the pouch contents shall be poured into a flat-bottom container. A minimum of three times the pouch's volume of 180°F to 190°F water shall be added to the container so as to cover the contents. The contents and water shall be gently agitated so as to liquefy rendered fat and to remove the sauce without breaking the beef dices or mushrooms. The contents shall then be poured into a U.S. Standard No. 7 sieve in a manner that will distribute the product over

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the sieve without breaking the beef dices or mushrooms. The sieve area shall be such that the distributed product does not completely cover all the openings of the sieve. The sieve shall be tilted at approximately a 45° angle and allowed to drain for two minutes before determining the drained weight by subtracting the sieve tare weight from the gross weight. The drained weight of the beef and mushrooms shall be determined individually and shall be reported to the nearest 0.1 ounce.

2/ If the sample average drained weight of beef is less than 2.5 ounces, the lot shall be rejected.

3/ If the sample average drained weight of mushrooms is less than 0.7 ounces, the lot shall be rejected.

4/ For purposes of this examination, the definition of a whole intact beef dice is one that measures not less than 1/4 inch in each of the two separate dimensions.

4.5.6 Fat and salt content testing. Eight filled and sealed pouches shall be selected at random from the lot. The product shall be tested for fat content in accordance with the Official Methods of Analysis of the AOAC, method 976.21, 985.15, or 960.39, and for salt content in accordance with the Official Methods of Analysis of the AOAC, method 935.47, except that preparation of the samples shall be as follows: The unopened pouches shall be gently warmed in a water bath to melt fat adhering to the inside of the pouches. The pouches shall be opened and the entire contents of each pouch shall be separately blended in a Waring Blendor or equivalent. The test results shall be reported to the nearest 0.1 percent. Any result failing to conform to the fat or salt requirements in 3.6 shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.7 Packaging examination. Examination of packaging shall be in accordance with the quality assurance provisions of MIL-P-44073.

5. PACKAGING

5.1 Packaging. Packaging shall be in accordance with the requirements of MIL-P-44073.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

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b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required (see 3.1, 4.4, and 6.2).
d. Provisions for approved preproduction samples (see 3.6.1 and 6.2).

6.2 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of Federal Acquisition Regulation (FAR) 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.3 Ingredient information.

6.3.1 Flavoring, braised beef. It has been found that braised beef flavoring (Bratarom) manufactured by Haarman and Reimer, Springfield, NJ 07081 meets the requirements of 3.2.11 and performs satisfactorily in this product.

6.4 Subject term (key word) listing.

Combat field feeding
Food processing
Operational ration
Pouch food
Shelf stable

Custodians:

Army - GL
Navy - SA
Air Force - 50

Review activities:

Army - MD, QM
Navy - MC
DLA - SS

Preparing activity:

Army - GL

(Project 8940-0747)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-B-44474	1993 September 20

3. DOCUMENT TITLE

BEEF WITH MUSHROOMS IN SAUCE, THERMOSTABILIZED, FOR MEAL, READY-TO-EAT

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S. Army Natick RD&E Center	(1) Commercial 508-651-4501	(2) AUTOVON/DSN 256-4501
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: SATNC-WT Natick, MA 01760-5018	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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